Methods of proof – 3

The extremal principle and compactness

Homework

13 сентября 2026 года

Theoretical minimum. The extremal principle is one line long: take the object that maximises something, and use that it cannot be improved. Choosing the functional is the whole difficulty, and the three usual choices are a size, a count and a distance. On infinite families the existence of the extremum has to be earned: through compactness and continuity, through a convergent subsequence, through Arzelà–Ascoli for families of functions, or through Zorn's lemma when no topology is available. The variational method is the same idea used constructively: maximise a functional and read off the conditions at the maximiser. If the supremum is not attained, either the family must be enlarged so that it becomes compact, or the limit object must be shown to be admissible — and this check is precisely the step that most wrong solutions omit.

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1. Prove that among all triangles inscribed in a given convex body there is one of maximal area, and that its vertices have the property that the tangent lines are parallel to the opposite sides.

2. Prove that a finite set of $n \geqslant 5$ points in the plane, no three collinear, has a subset of four points forming a convex quadrilateral, by taking the convex hull and arguing extremally (compare with Convex – 1).

3. Let f be continuous on $\mathbb{R}$ with $f(x) \to 0$ as $|x| \to \infty$ and f not identically zero. Prove that f attains either its maximum or its minimum, and give an example where only one of the two is attained.

4. Prove that every bounded infinite subset of $\mathbb{R}^n$ has an accumulation point, and deduce that a sequence of nested non-empty compact sets

has a common point; then give an example of nested closed unbounded sets with empty intersection.

5. Prove that in a finite metric space there exist two points at maximal distance, and that this pair is not necessarily unique; then prove that a compact metric space has a pair realising its diameter.

6. Prove the existence of a best uniform approximation to a continuous function by polynomials of degree at most n, and of a best L^2 approximation, and prove that both are unique, explaining why the two uniqueness proofs are different (compare with Continuous functions – 9).

7. Prove that a non-constant polynomial attains its minimum modulus on $\mathbb{C}$, and deduce the fundamental theorem of algebra by showing that the minimum cannot be non-zero (d'Alembert's argument).

8. Let $\mathcal{A}$ be a family of open intervals covering $[0,1]$. Prove by the extremal principle that a finite subfamily covers $[0,1]$, and compare this with the standard compactness proof.

9. Prove that among all closed curves of a given length the circle encloses the largest area, at least in the class of convex curves, by a compactness plus symmetrisation argument, and say where the existence of the maximiser is used (Convex – 4).

10. Let T be an infinite tree in which every vertex has finite degree. Prove that T has an infinite path (Kőnig's lemma), and deduce the compactness principle for propositional logic in the countable case.